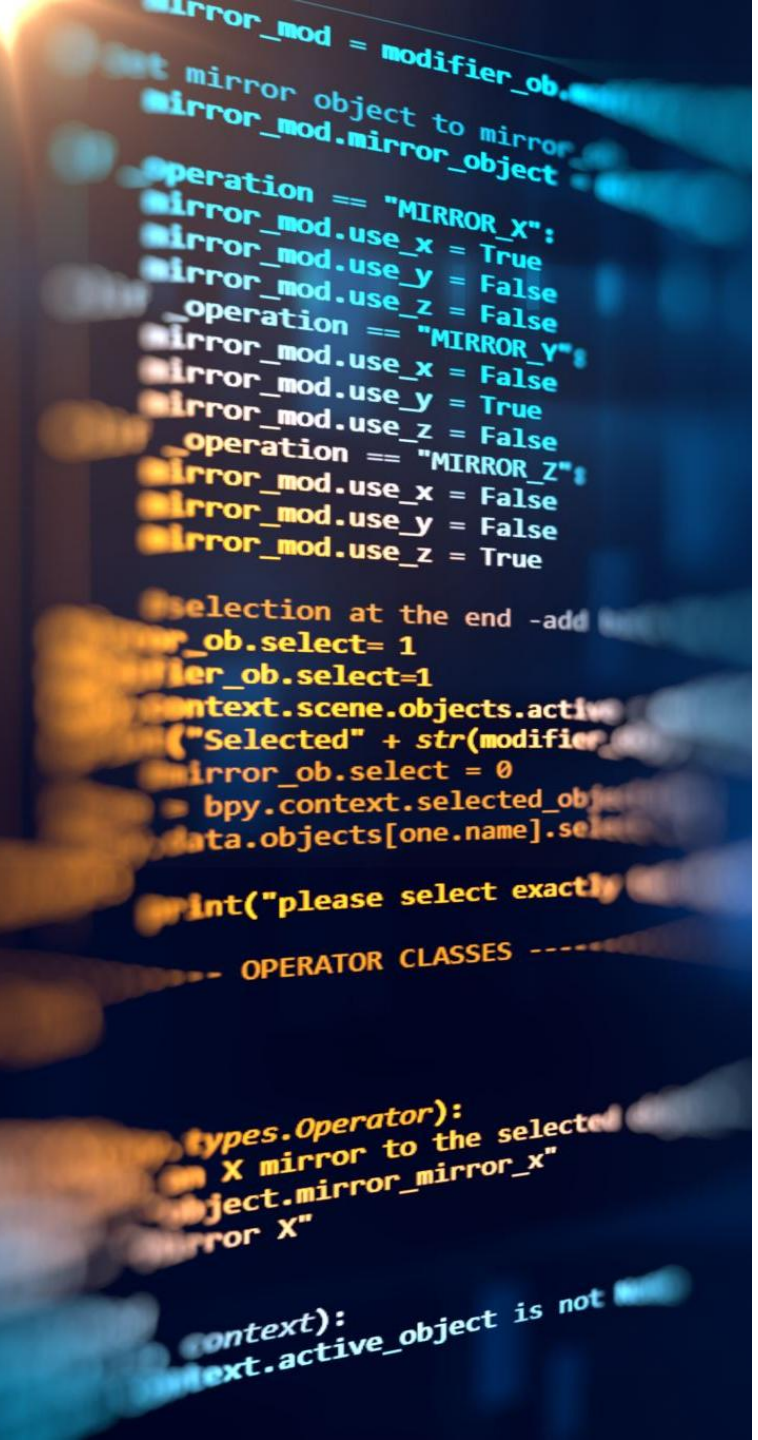
Spring of Code

Artificial Intelligence

(Machine Learning & Deep Learning)

Instructor
Asad Ullah
- *Research Assistant* (DIP Lab)

Duration
03 Months
(Feb – April)



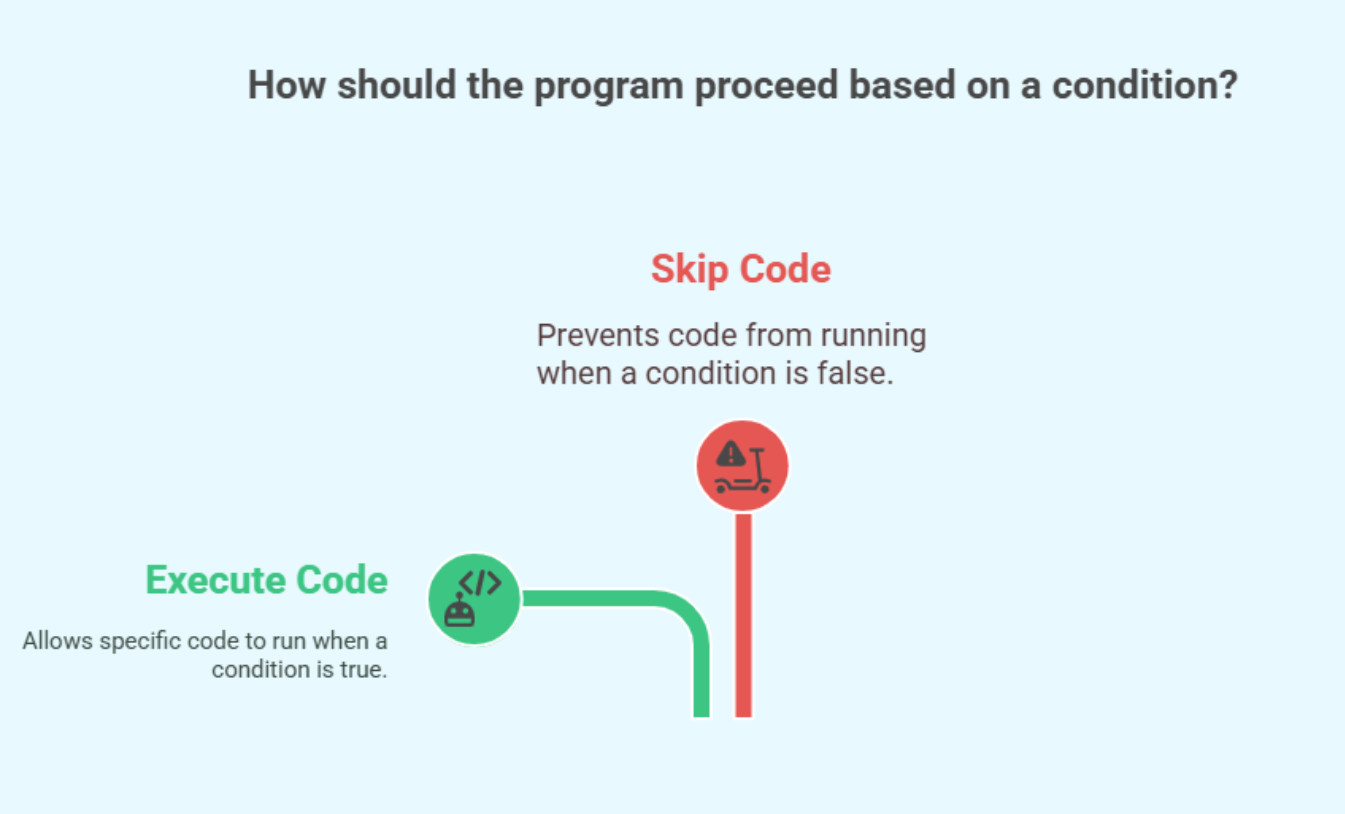
Day 05 – Python Fundamentals (Conditionals and Loops)

Objectives:

- What are Conditional Statements?
- What are Loops?
- Python Indentation
- What are Iterables?

Conditional Statements

- **What Are Conditional Statements?**
  - Conditional statements allow a program to make decisions based on whether a condition is **True** or **False**.
- **Why Use Conditions?**
  - Control the flow of execution.
  - Executes code only when specific conditions are met.
  - Skip or branch logic based on outcomes.
- **Real-Life Example**: Traffic light
  - If the light is green → cars go.
  - If the light is red → cars stop.



Conditions in Python

- A condition is an expression that evaluates to True or False.

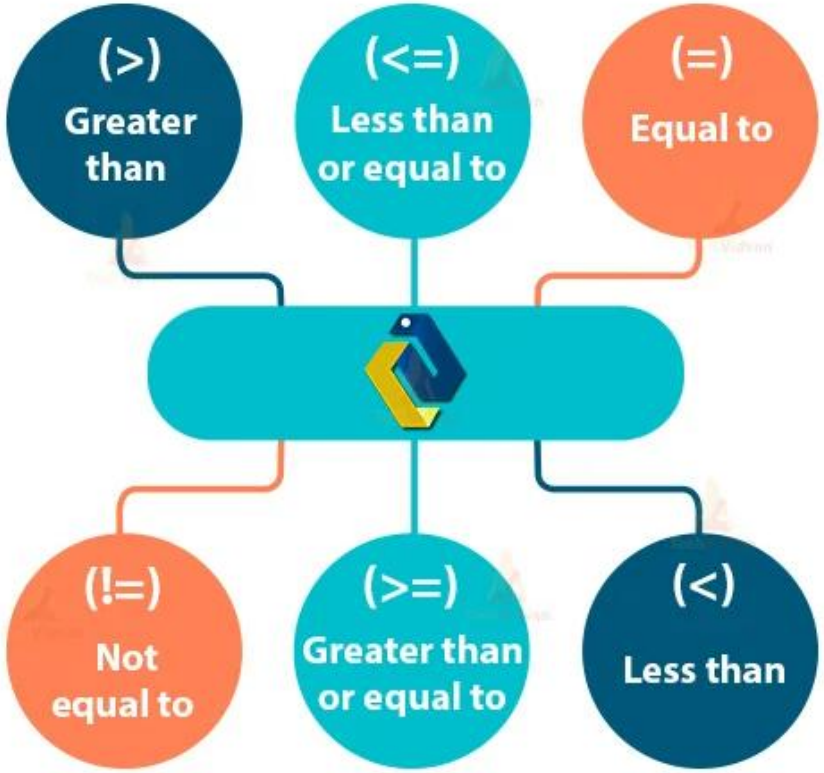
Common Comparison Operators:

Operator	Meaning
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

Example:

```
x = 10
if x > 5:
    print("x is greater than 5")
```

- In this example, ***x > 5*** is the condition being evaluated.

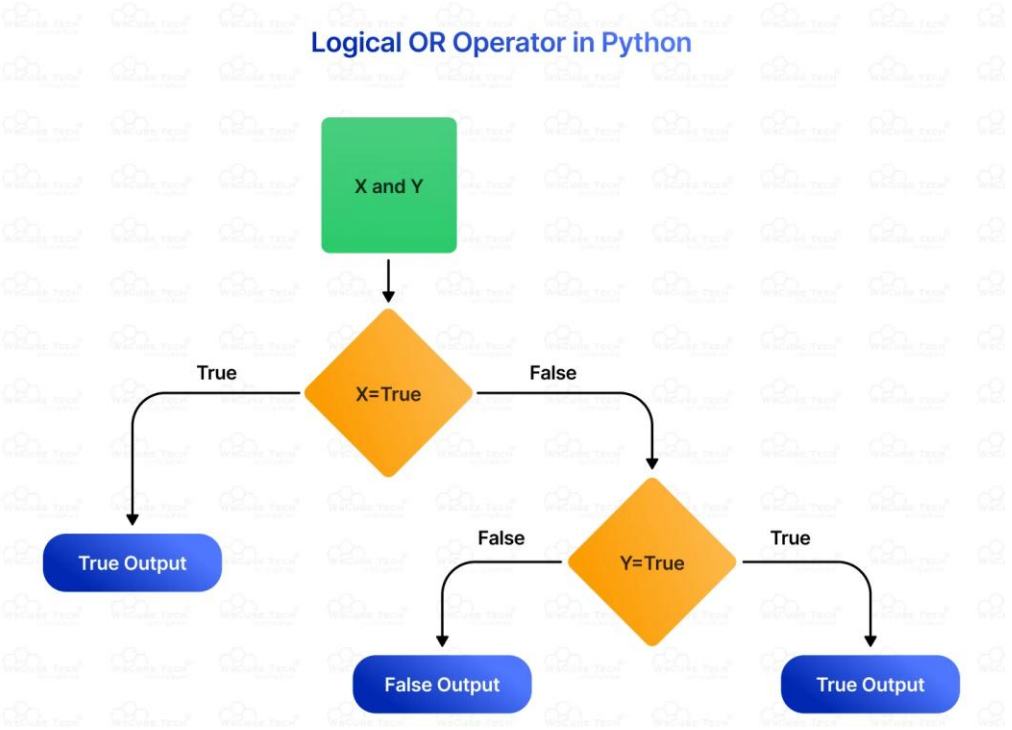


Logical Operators in Python

Logical operators are used to combine multiple conditions and return a True or False result.

Types of Logical Operators:

Operator	Description
and	True if both conditions are True
or	True if at least one condition is True
not	Reverses the result (True ↔ False)



Examples:

```
# and operator
age = 20
if age > 18 and age < 30:
   print("You're a young adult")

# or operator
grade = 'B'
if grade == 'A' or grade == 'B'
     print("You passed")
```

if Statements

Purpose:

- if statements allow you to execute code only when a specific condition is True.

Syntax:

```
if condition:
    # code block to execute if condition is True
```

- The condition follows the if keyword and ends with a colon :.
- The code block is indented and runs only if the condition is True.
- If the condition is False, the block is skipped.

Example:

```
num = 10
if num > 5:
    print("The number is greater than 5")
```

- In this example, the message is printed only if *num > 5*

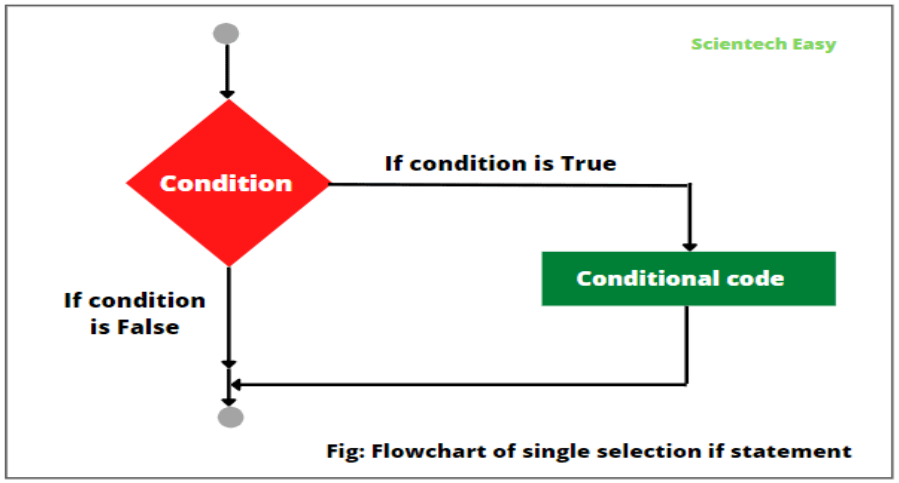


Fig: Flowchart of single selection if statement

```
name = 'Jason'
if name == 'Jason':
    print("Hello Jason, Welcome")
else:
    print("Sorry, I don't know you")
```

if-else Statements

Purpose:

- if-else statements allow a program to take two different actions based on whether a condition is True or False

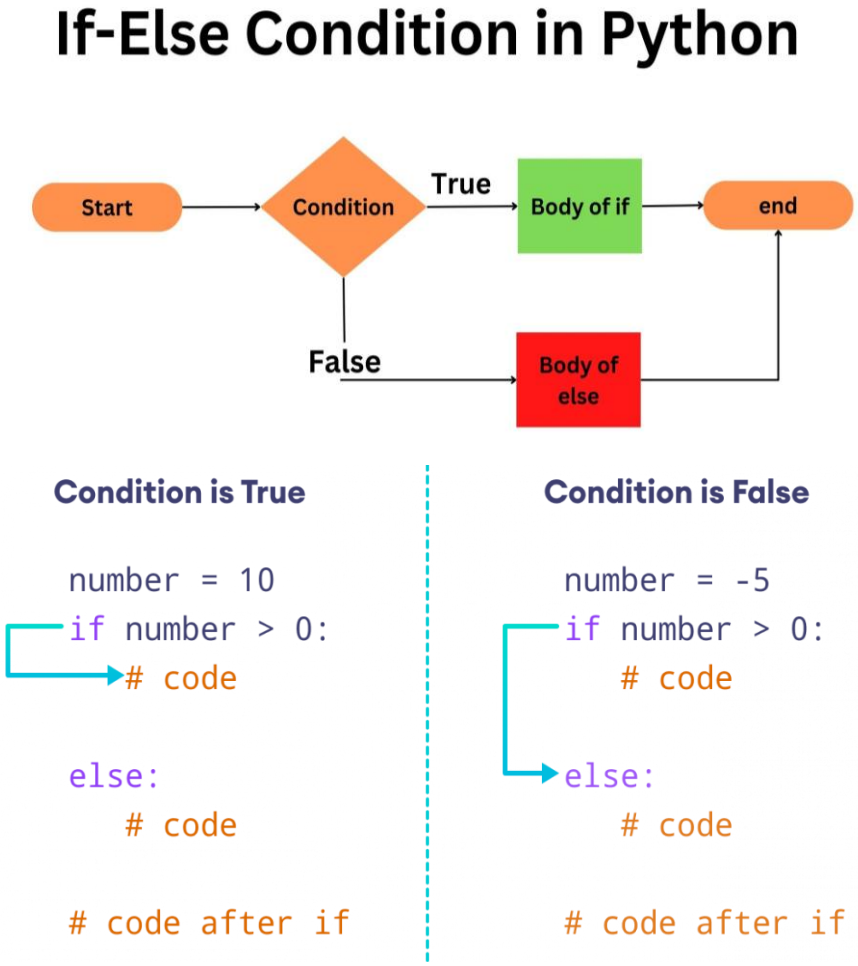
How it Works:

- The **if** block executes when the condition is True.
- The **else** block executes when the condition is False.
- This ensures that one of the two blocks will always run.

Example:

```
num = 3
if num > 5:
    print("The number is greater than 5")
else:
    print("The number is not greater than 5")
```

- Here, since *num* is not greater than 5, the else block runs and prints the second message



if-elif-else Statements

Purpose:

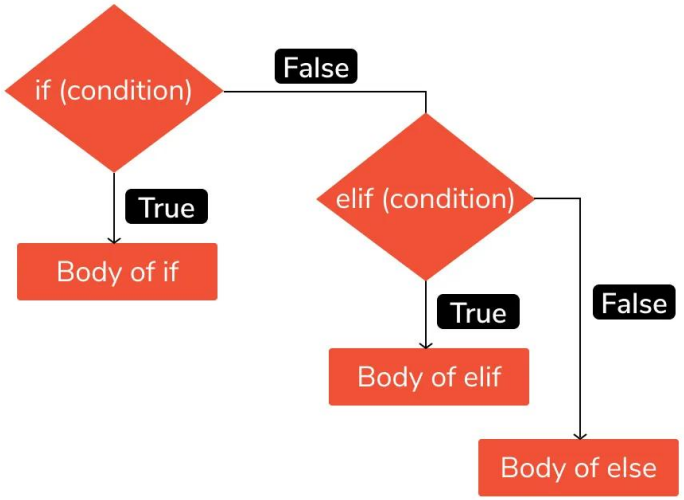
- Used when you need to evaluate multiple conditions and execute different blocks of code based on which condition is True.

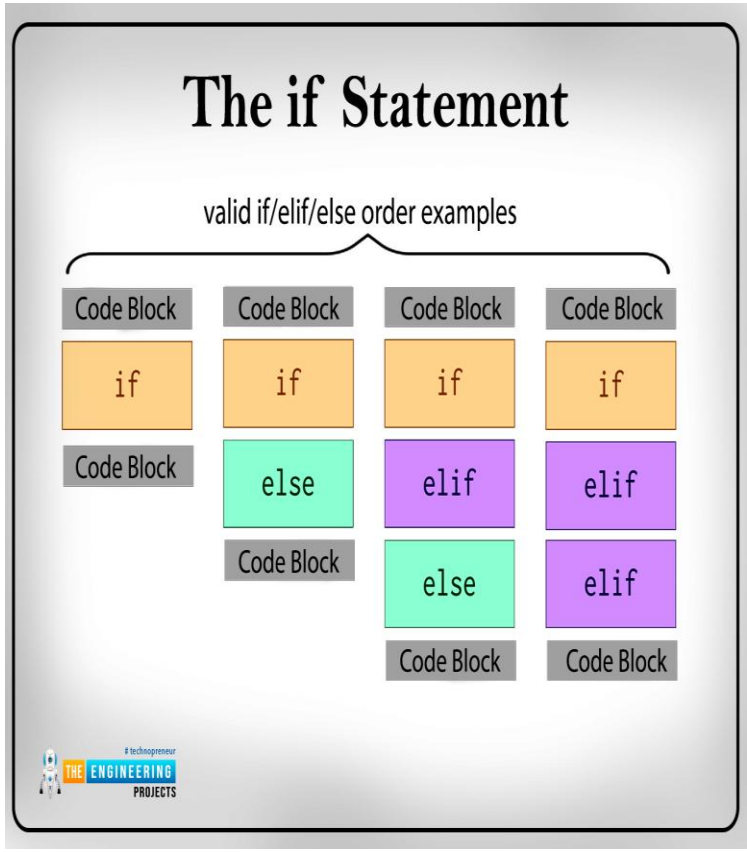
How it Works:

- The program checks the first condition with if.
- If True, it runs that block and skips the rest.
- If False, it checks the next condition using elif (else if).
- You can have multiple elif blocks.
- If none of the conditions are True, the else block is runs as a fallback.

- **Example**

```
score = 75
if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
elif score >= 70:
    print("Grade: C")
else:
    print("Grade: D")
```





Introduction to Loops

Definition:

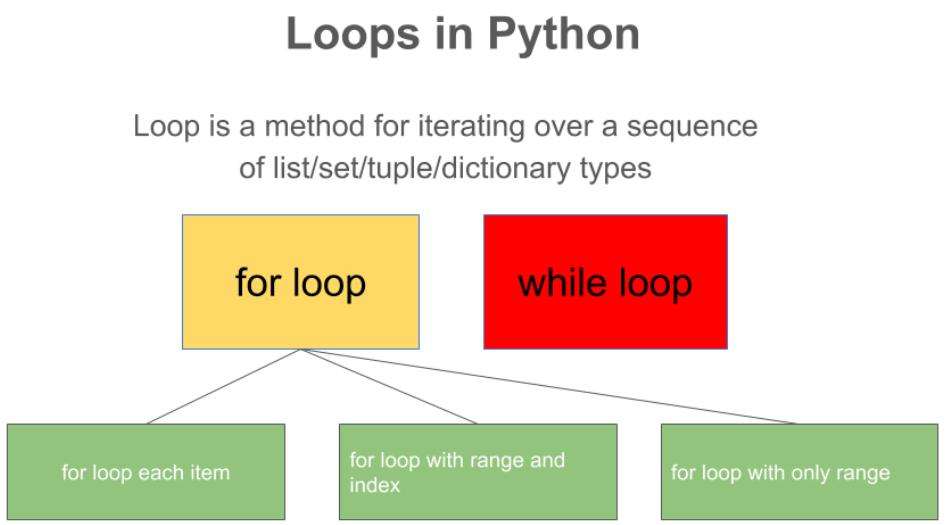
- Loops are control structures used to repeat a block of code until a specific condition is met.

Purpose:

- Automate repetitive tasks and reduce redundancy.

Types of Loops:

- **while Loop**: Repeats as long as a condition is true.
- **for Loop**: Iterates over items of a sequence like lists, tuples, or strings.

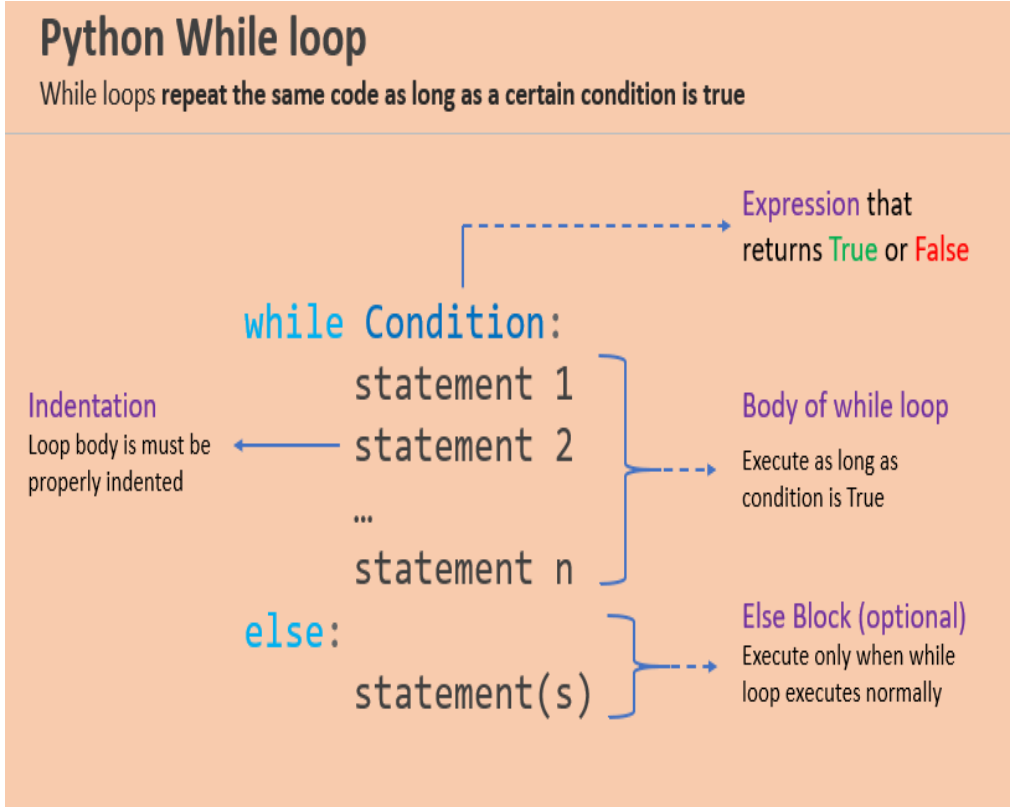


while Loop

- **How it Works:**
  - The while loop runs as long as the condition is true. If the condition becomes false, the loop stops.
- **Use Case:**
  - Ideal when the number of iterations is not known in advance.
- **Example:**

```
count = 0
while count < 3:
    print("Counting:", count)
    count += 1
```

- The loop checks if *count* is less than 3. If true, it prints the value and increases *count* by 1.
- Prints Counting: 0, Counting: 1, Counting: 2. Stops when count == 3.



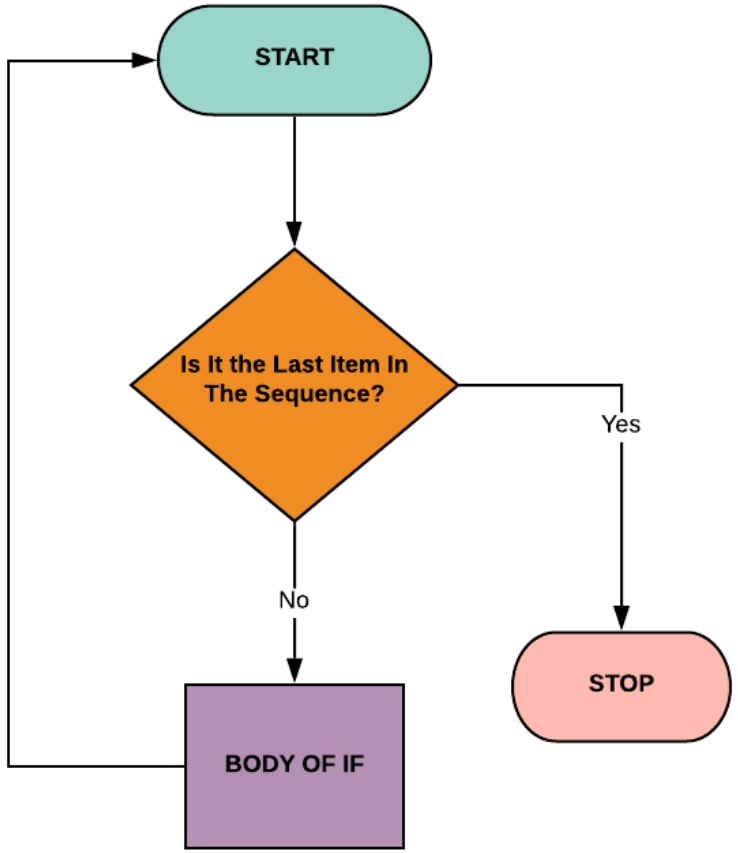
for Loop

- **How it Works:**
  - The for loop iterates items in a sequence (list, tuple, string, etc.).
- **Use Case:**
  - Best when the number of iterations is know or fixed.
- **Example 1 – List:**

```
colors = ["red", "green", "blue"]
for color in colors:
    print("Color:", color)
```

- **Example 2 – Dictionary:**

```
data = {"name": "John", "age": 25}
for key, value in data.items():
    print(f"{key}: {value}")
```



range() Function

- Generates a sequence of numbers, commonly used with for loops.

- **Syntax:**

  range(start, stop, step)

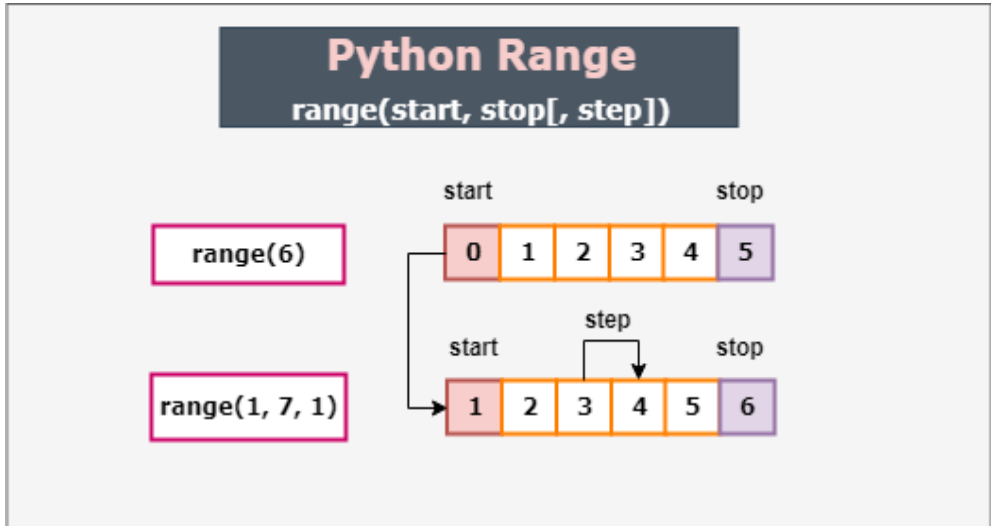
- **Parameters of range():**
  - **start:** Starting number (default is 0).
  - **stop:** End number (non-inclusive).
  - **step:** Increment (default is 1).

- **Examples:**
  - **Standard Loop:**

```
for i in range(1, 6):
    print(i)  # Prints 1 to 5
```

  - **Reverse Loop:**

```
for i in range(5, 0, -1):
    print(i)  # Prints 5 to 1
```



break Statement

- **Purpose:**
  - Immediately stops the loop when a condition is met.
- **Example:**

```
for number in range(10):
    if number == 5:
        break
    print(number)
```

- Prints 0 to 4. Stops at 5.

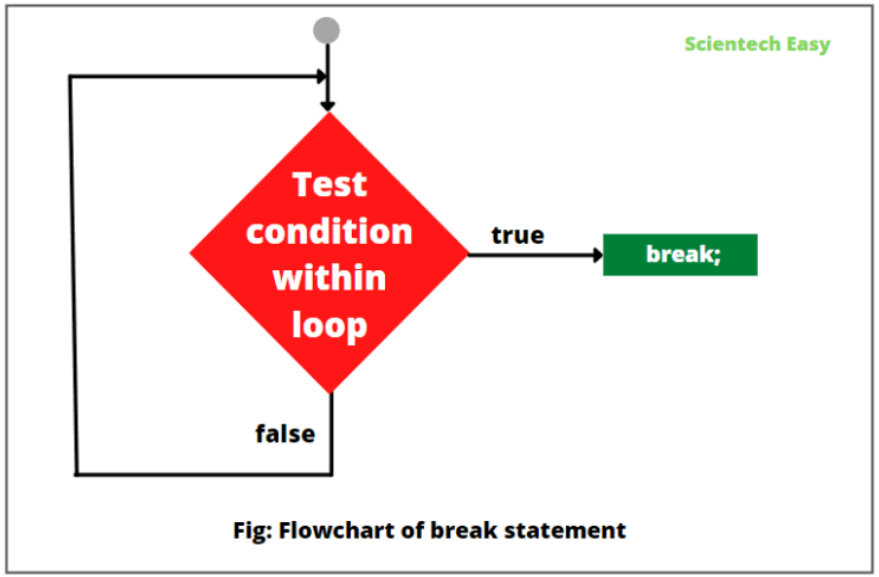


Fig: Flowchart of break statement

```
for val in sequence:
  # code
  if condition:
    break

  # code
```

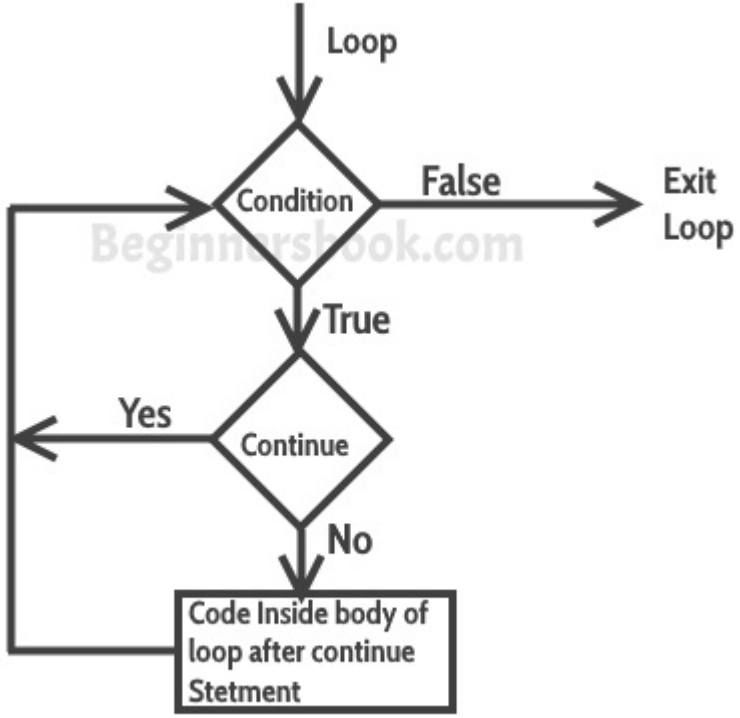
```
while condition:
  # code
  if condition:
    break

  # code
```

continue statement

- Skips the current iteration and moves to the next one.
- It doesn't terminate the loop but simply skips the remaining code for that iteration.
- **When to Use continue:**
  - To skip unwanted iterations (e.g., skipping certain values in a dataset).
- **Example:**

```
for i in range(5):
    if i == 2:
        continue
    print(i)
```

- Skips 2. Prints 0, 1, 3, 4.




```
students = ['Ashton', 'Jack', 'Rose', 'Tim', 'Elle', 'Johnny', 'Sammy',
    'David', 'Monica', 'Arjun']

for n in students:
    if len(n) == 4:
        continue
    print('Hello', n)
```

pass statement

- **Purpose:**
  - Does nothing, used where code is syntactically required but not yet implemented.
  - The pass statement is a null operation; nothing happens when it is executed. It's used as a placeholder for code you'll add later.
- **Example with pass:**

```
for i in range(5):
    if i < 3:
        pass
print(i)
```

- *pass* does nothing; all numbers 0 to 4 are printed.

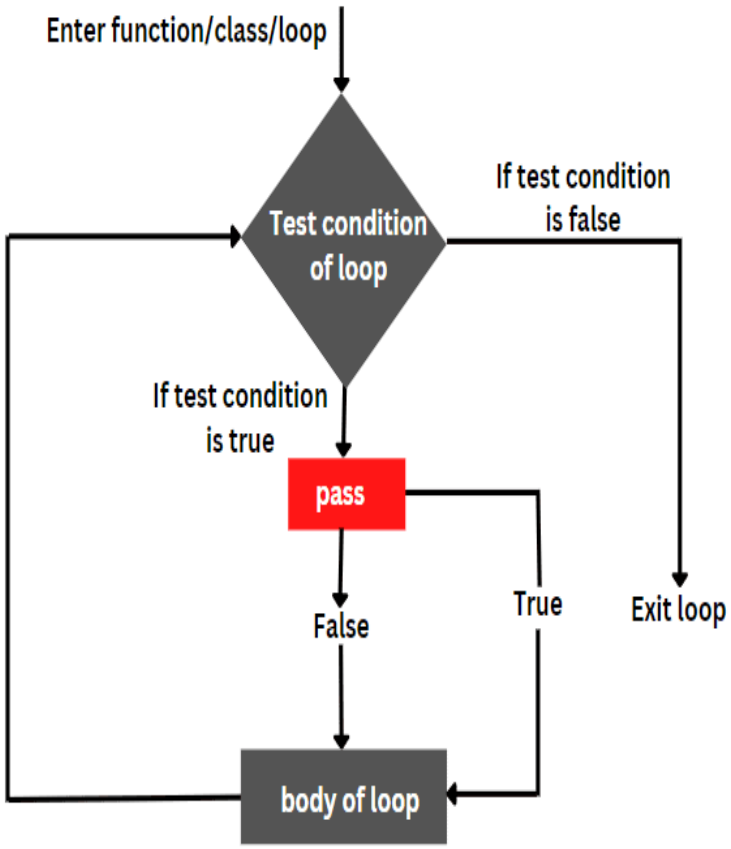


Fig: Flowchart diagram of Pass statement in Python

python_code.py > ...

```
i = 1

if(i <= 10):
    pass

print("outside if statement")
```

Happy Coding